

AIMLCZG521 — Conversational AI

Group Assignment 1 — Problem Statement 2

Study of Embedding Models and Approximate Nearest Neighbor Search:

Semantic Quality vs Search Efficiency

Group 129

Student Details

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Group Contributions

Student	Contribution Area	Specific Contribution
B Devchandraaj	Dataset & embedding	Dataset loading, validation, and embedding generation
Pushadapu Sanjay Kumar	Similarity & exact kNN	Metric formulas, normalization analysis, and exact retrieval
Gourob Nandi	HNSW & IVF	FAISS index construction and ANN parameter sweeps
Anirudh Anand	Evaluation & reporting	Metrics, visualizations, qualitative examples, and report preparation

Executive Summary

Objective. Compare two encoder embedding models and two approximate-nearest-neighbour (ANN) indexes on a real retrieval task, trading off semantic quality vs. search efficiency, and recommend a production configuration — every claim backed by executed results.

Dataset. SciFact (BEIR benchmark): 5,183 document passages, 300 queries each with relevance judgments (339 total qrels). Clears the assignment's 1,000-passage / 50-query minimums by ~5x and ~6x respectively.

Models. distilbert-base-uncased (768-dim, mean pooling, general-purpose) vs. BAAI/bge-large-en-v1.5 (1024-dim, [CLS] pooling, retrieval-tuned) — a deliberately contrasting pair.

Key Results (executed)

Finding	Result
Best embedding model (exact Recall@5)	BGE-large 0.780 vs DistilBERT 0.108 (~7.2x higher)
Most appropriate metric	Cosine ? identical ranking to dot/L2 on unit-normalized vectors; scale-invariant
Best ANN trade-off	HNSW efSearch=16 - Recall@5 0.770, 0.077 ms/query, 1.45x faster than exact; fastest eligible config clearing the 0.75 bar
Fastest ANN (rejected by quality bar)	IVF nprobe=1 - fastest overall but Recall@5 only 0.502 (below the quality bar)
Model disagreement (Task 7)	Mean Top-5 set overlap 0.135 — the two models retrieve largely different documents

Recommendation. Embed with BGE-large (unit-normalized) and serve HNSW efSearch=16 — the fastest eligible configuration in the executed run (Recall@5 0.770, 0.077 ms/query, 1.45x faster than exact), clearing the Recall@5 ≥ 0.75

quality bar. Where RAM is the binding constraint, use IVF nprobe=20 (Recall@5 0.769, 0.166 ms/query, 0.68x versus exact) and verify the footprint on target hardware.

Problem Statement

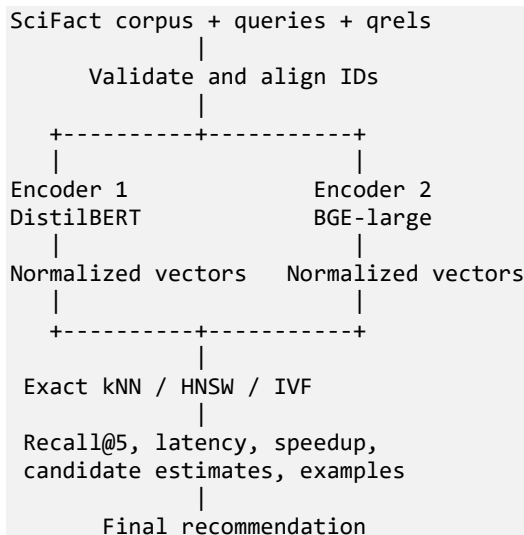
This assignment studies semantic retrieval quality and approximate-nearest-neighbor search efficiency using two embedding models, across four modules: (1) dataset and embedding preparation, (2) similarity metrics and exact retrieval, (3) HNSW and IVF ANN search, and (4) embedding quality analysis and final recommendation.

Tools and Libraries Used

- Python 3.14.5 (recorded execution environment)
- datasets (Hugging Face) — loading the BEIR-format SciFact dataset
- pandas, numpy — data handling
- sentence-transformers — encoder embedding models
- faiss-cpu — HNSW / IVF ANN indexes
- matplotlib — plots

Experimental Methodology

The same corpus, query set, document/query identifiers, normalized vectors, and qrels are used for every comparison. Documents are encoded offline; queries are encoded at search time. Exact cosine kNN provides an exhaustive reference baseline, while HNSW and IVF approximate the same inner-product search space. Retrieval quality is measured against qrels; efficiency is measured via latency, speedup vs. exact search, and candidate-vector estimates.

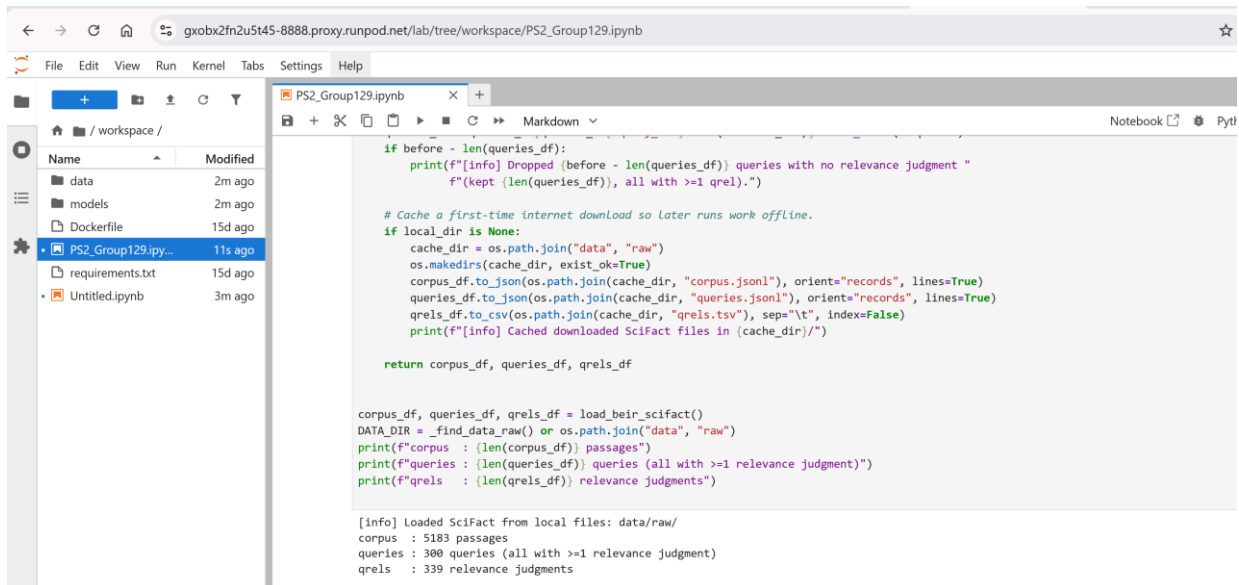


Task 1 — Dataset Details and Source

- Name: SciFact (BEIR benchmark; biomedical / scientific-claim verification)
- Citation: Wadden et al., Fact or Fiction: Verifying Scientific Claims, EMNLP 2020; redistributed by Thakur et al., BEIR, NeurIPS 2021 <https://aclanthology.org/2020.emnlp-main.609/>
- Source: <https://huggingface.co/datasets/BeIR/scifact> (corpus + queries), <https://huggingface.co/datasets/BeIR/scifact-qrels> (relevance judgments)
- Size (executed): 5,183 corpus passages, 300 queries each with ≥ 1 relevance judgment, 339 total qrels, binary relevance

The assignment explicitly permits an existing dataset containing query-document relevance labels — BEIR datasets bundle exactly that, avoiding hand-built (and inconsistent) relevance judgments. SciFact's claim-verification domain gives clean,

unambiguous labels, useful later for Tasks 3 and 7. Corpus and queries clear the assignment minimums by ~5x and ~6x respectively.



```
if before - len(queries_df):
    print(f"[info] Dropped {before - len(queries_df)} queries with no relevance judgment "
          f"({kept {len(queries_df)}, all with >=1 qrel}).")

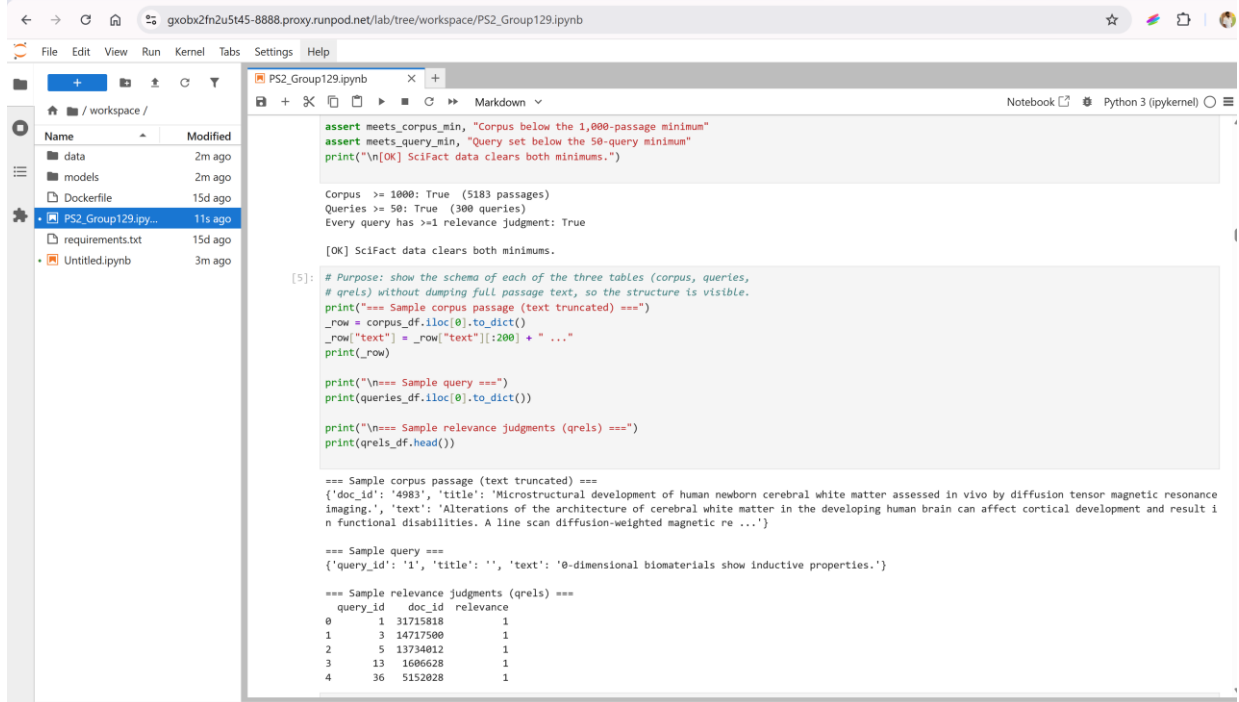
# Cache a first-time internet download so later runs work offline.
if local_dir is None:
    cache_dir = os.path.join("data", "raw")
    os.makedirs(cache_dir, exist_ok=True)
    corpus_df.to_json(os.path.join(cache_dir, "corpus.jsonl"), orient="records", lines=True)
    queries_df.to_json(os.path.join(cache_dir, "queries.jsonl"), orient="records", lines=True)
    qrels_df.to_csv(os.path.join(cache_dir, "qrels.tsv"), sep="\t", index=False)
    print(f"[info] Cached downloaded SciFact files in {cache_dir}/")

return corpus_df, queries_df, qrels_df

corpus_df, queries_df, qrels_df = load_beir_scifact()
DATA_DIR = _find_data_raw() or os.path.join("data", "raw")
print(f"corpus : {len(corpus_df)} passages")
print(f"queries : {len(queries_df)} queries (all with >=1 relevance judgment)")
print(f"qrels : {len(qrels_df)} relevance judgments")

[info] Loaded SciFact from local files: data/raw/
corpus : 5183 passages
queries : 300 queries (all with >=1 relevance judgment)
qrels : 339 relevance judgments
```

Task 1 Virtual Lab execution evidence: SciFact corpus, query, and qrels counts after dataset loading.



```
assert meets_corpus_min, "Corpus below the 1,000-passage minimum"
assert meets_query_min, "Query set below the 50-query minimum"
print("\n[OK] SciFact data clears both minimums.")

Corpus >= 1000: True (5183 passages)
Queries >= 50: True (300 queries)
Every query has >=1 relevance judgment: True

[OK] SciFact data clears both minimums.

[5]: # Purpose: show the schema of each of the three tables (corpus, queries,
# qrels) without dumping full passage text, so the structure is visible.
print("=== Sample corpus passage (text truncated) ===")
_row = corpus_df.iloc[0].to_dict()
_row["text"] = _row["text"][:200] + " ..."
print(_row)

print("\n=== Sample query ===")
print(queries_df.iloc[0].to_dict())

print("\n=== Sample relevance judgments (qrels) ===")
print(qrels_df.head())

=== Sample corpus passage (text truncated) ===
{'doc_id': '4983', 'title': 'Microstructural development of human newborn cerebral white matter assessed in vivo by diffusion tensor magnetic resonance imaging.', 'text': 'Alterations of the architecture of cerebral white matter in the developing human brain can affect cortical development and result in functional disabilities. A line scan diffusion-weighted magnetic re ...'}

=== Sample query ===
{'query_id': '1', 'title': '', 'text': '0-dimensional biomaterials show inductive properties.'}

=== Sample relevance judgments (qrels) ===
query_id doc_id relevance
0 1 31715818 1
1 3 14717500 1
2 5 13734012 1
3 13 1606628 1
4 36 5152028 1
```

Task 1 Virtual Lab execution evidence: resolved SciFact files saved to the local data/raw cache.

```

[6]: # Purpose: write corpus/queries/qrels to the resolved data/raw
# directory (DATA_DIR, set in the Load cell above) so Tasks 2+ can load the
# exact same fixed dataset without re-downloading or re-running this cell.
import os

# Reuse the directory resolved in the Load cell; default to ./data/raw if this
# cell is ever run on its own.
DATA_DIR = globals().get("DATA_DIR", os.path.join("data", "raw"))
os.makedirs(DATA_DIR, exist_ok=True)

corpus_df.to_json(os.path.join(DATA_DIR, "corpus.jsonl"), orient="records", lines=True)
queries_df.to_json(os.path.join(DATA_DIR, "queries.jsonl"), orient="records", lines=True)
qrels_df.to_csv(os.path.join(DATA_DIR, "qrels.tsv"), sep="\t", index=False)

print(f"Saved to {DATA_DIR}/: corpus.jsonl, queries.jsonl, qrels.tsv")

```

Saved to data/raw/: corpus.jsonl, queries.jsonl, qrels.tsv

Task 1 Virtual Lab execution evidence: sample corpus passage, query, and relevance judgments.

Task 2 — Embedding Generation and Pooling

Model	Dimension	Max input length	Pooling	Approx. doc embedding time
distilbert-base-uncased	768	512 tokens	Mean pooling	341.12 s (CPU)
BAAI/bge-large-en-v1.5	1024	512 tokens	[CLS] token pooling	2034.55 s (CPU)

Task 2 was executed in the RunPod Virtual Lab using the CPU configuration recorded in the notebook (batch size 32 and 12 CPU threads). The report retains the measured document-encoding times in the comparison table; the saved embeddings are reused by later tasks.

```

max_seq_length = spec["max_input_length"]
"parallel_models": False, "parallel_batches": False,
"batch_workers": 1, "chunk_size": None)

json.dump(timings, open(TIMINGS_PATH, "w", encoding="utf-8"), indent=2)
model_records[model_name] = {"elapsed": result["documents_s"]}
print(f"Generated + saved: docs={doc_emb.shape}, queries={qry_emb.shape}, "
      f"time={result['documents_s']:.1f}s")

comparison_rows = []
for model_name, spec in MODEL_SPECS.items():
    elapsed = model_records.get(model_name, {}).get("elapsed")
    comparison_rows.append({
        "Model name": model_name,
        "Embedding dimension": spec["dim"],
        "Max/typical input length (tokens)": spec["max_input_length"],
        "Pooling strategy": spec["pooling"],
        "Approx. document embedding time (s)": round(float(elapsed), 2) if elapsed is not None else None,
        "Timing status": "Executed" if elapsed is not None else "Execution required before submission",
    })

comparison_df = pd.DataFrame(comparison_rows)
comparison_df

[Task 2] device=cpu, batch_size=32, cpu_threads=12
=== distilbert-base-uncased ===
Loaded saved embeddings: docs=(5183, 768), queries=(300, 768)
=== BAAI/bge-large-en-v1.5 ===
Loaded saved embeddings: docs=(5183, 1024), queries=(300, 1024)

```

	Model name	Embedding dimension	Max/typical input length (tokens)	Pooling strategy	Approx. document embedding time (s)	Timing status
0	distilbert-base-uncased	768	512	mean pooling	341.12	Executed
1	BAAI/bge-large-en-v1.5	1024	512	[CLS] token pooling	2034.55	Executed

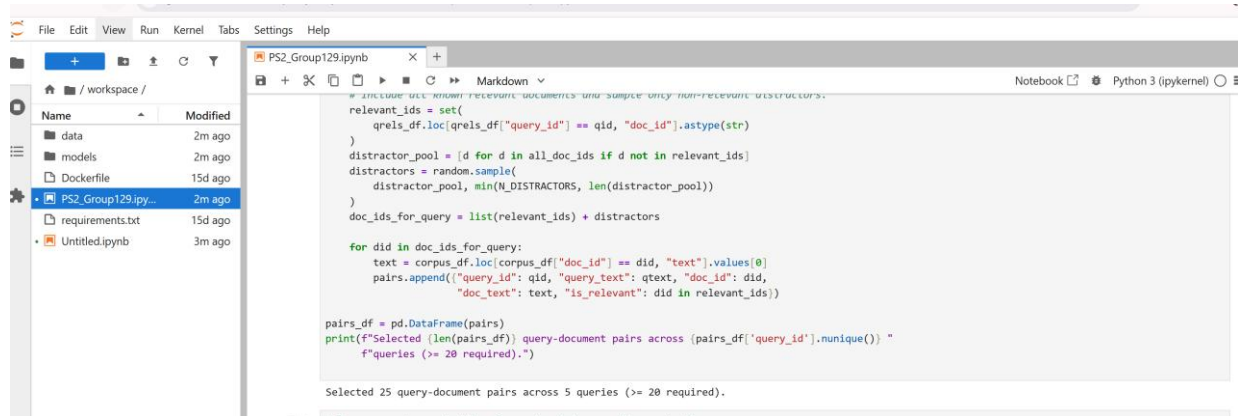
Task 2 Virtual Lab execution evidence: CPU timing and embedding shapes.

Encoder models are appropriate for semantic representation because their bidirectional attention produces context-aware token representations that, when pooled into a single fixed-length vector, place semantically similar texts near each other in vector space — enabling similarity search via simple distance metrics instead of exact keyword matching. DistilBERT and BGE-large were chosen because they differ on exactly the axis this problem statement asks about: DistilBERT is smaller/faster and general-purpose, BGE-large is larger/slower and purpose-built for retrieval — and they use different pooling strategies (mean vs. [CLS]), giving Task 7 a real, explainable axis of disagreement.

Module 2 — Similarity and Exact Retrieval

Task 3 — Similarity Metric Comparison

25 query-document pairs were sampled across 5 queries (≥ 20 required): each query paired with its known-relevant documents plus 4 random distractors. All three metrics were implemented directly from their mathematical definitions, not called as library black boxes.



```
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models 2m ago
Dockerfile 15d ago
PS2_Group129.py... 2m ago
requirements.txt 15d ago
Untitled.ipynb 3m ago

> INCLUDE ALL KNOWN RELEVANT DOCUMENTS AND SAMPLE ONLY NON-RELEVANT DISTRACTORS.
relevant_ids = set(
    qrels_df.loc[qrels_df["query_id"] == qid, "doc_id"].astype(str)
)
distractor_pool = [d for d in all_doc_ids if d not in relevant_ids]
distractors = random.sample(
    distractor_pool, min(N_DISTRACTORS, len(distractor_pool))
)
doc_ids_for_query = list(relevant_ids) + distractors

for did in doc_ids_for_query:
    text = corpus_df.loc[corpus_df["doc_id"] == did, "text"].values[0]
    pairs.append(("query_id": qid, "query_text": qtext, "doc_id": did,
                "doc_text": text, "is_relevant": did in relevant_ids))

pairs_df = pd.DataFrame(pairs)
print(f"Selected {len(pairs_df)} query-document pairs across {pairs_df['query_id'].nunique()} "
      f"queries (>= 20 required).")

Selected 25 query-document pairs across 5 queries (>= 20 required).
```

Task 3 Virtual Lab execution evidence: metric definitions and pair selection.

Finding 1 — BGE-large (already unit-normalized)

Document-vector norms: mean=1.0000, std=0.0000. Ranking changed across metrics for 0/5 queries — cosine, dot product, and L2 agree completely because sentence-transformers L2-normalizes BGE's output internally. This demonstrates the identities $\text{dot}(\hat{a}, \hat{b}) = \cos(a, b)$ and $\|\hat{a} - \hat{b}\|^2 = 2 - 2 \cos(a, b)$ hold for unit vectors, but shows no divergence to explain.

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PS2_Group129.ipynb

Reused saved BAAI/bge-large-en-v1.5 embeddings for 25 pairs.

	query_id	doc_id	is_relevant	cosine	dot_product	L2_distance
0	660	1215116	True	0.613863	0.613863	0.878792
1	660	41496215	False	0.407035	0.407035	1.089004
2	660	32177659	False	0.447672	0.447672	1.051026
3	660	13702924	False	0.500359	0.500359	0.999641
4	660	2048139	False	0.525342	0.525342	0.974329
5	1362	8290953	True	0.584091	0.584090	0.912041
6	1362	22025252	False	0.489939	0.489939	1.010011
7	1362	154763124	False	0.441601	0.441601	1.056787
8	1362	10795340	False	0.477080	0.477080	1.022663
9	1362	17119869	False	0.498330	0.498330	1.001669
10	185	18340282	True	0.674860	0.674860	0.806399
11	185	11335860	False	0.531729	0.531729	0.967751
12	185	4702639	False	0.547632	0.547632	0.951176
13	185	36288526	False	0.444918	0.444918	1.053644
14	185	6826636	False	0.481996	0.481996	1.017845
15	743	32159283	True	0.776836	0.776836	0.668078
16	743	6082738	False	0.517617	0.517617	0.982225
17	743	13798951	False	0.510560	0.510560	0.989383
18	743	38784540	False	0.501756	0.501756	0.998242
19	743	11291348	False	0.530083	0.530083	0.969450
20	1359	11614737	True	0.866433	0.866433	0.516850
21	1359	6144969	False	0.450090	0.450090	1.048723

Task 3 Virtual Lab execution evidence: 25 query-document pairs and similarity values.

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PS2_Group129.ipynb

Raw document embedding norms: mean=1.0000, std=0.0000 (dot product can only disagree with cosine i

```

=== Ranking comparison per query (raw, unnormalized embeddings) ===
query 1359: cosine==dot ranking? True | cosine==L2 ranking? True
query 1362: cosine==dot ranking? True | cosine==L2 ranking? True
query 185: cosine==dot ranking? True | cosine==L2 ranking? True
query 660: cosine==dot ranking? True | cosine==L2 ranking? True
query 743: cosine==dot ranking? True | cosine==L2 ranking? True
Summary: ranking changed across metrics for 0/5 queries.

=== Normalization effect: recompute dot/L2 on unit-normalized embeddings ===
Summary: identity held for 25/25 pairs (dot) and 25/25 pairs (L2).

```

Task 3 Virtual Lab execution evidence: BGE ranking and normalization checks.

Finding 2 — DistilBERT (genuinely un-normalized)

Document-vector norms: mean=7.851, std=0.313 (range 7.42–8.56). Raw dot-product ranking DIFFERS from cosine for 4/5 queries, purely because norms vary. After L2-normalization, the identities hold exactly: dot==cosine for 25/25 pairs, $\|a-b\|^2=2-2\cdot\cos$ for 25/25 pairs.

```

print("Skipped -- needs metrics_df (cell above) and DistilBERT embeddings from Task 2.")

distilbert-base-uncased: raw document-vector norms mean=7.851, std=0.313 (range 7.42-8.56) -- genuinely un-normalized.

=== RAW (un-normalized) ranking comparison per query ===
query 1359: cosine==dot ranking? False | cosine==l2 ranking? True
query 1362: cosine==dot ranking? False | cosine==l2 ranking? False
query 185: cosine==dot ranking? False | cosine==l2 ranking? True
query 660: cosine==dot ranking? True | cosine==l2 ranking? False
query 743: cosine==dot ranking? False | cosine==l2 ranking? False
Summary: dot-product ranking DIFFERS from cosine for 4/5 queries -- purely because the raw vector norms vary.

=== After L2-normalization (unit vectors) ===
dot(a,b) == cosine(a,b) held for 25/25 pairs
||a-b||^2 == 2 - 2*cosine held for 25/25 pairs

Conclusion: when norms vary (DistilBERT) normalization changes the dot-product ranking to agree with cosine; when vectors are already unit-length (BGE) it has no effect. This is the practical reason retrieval systems normalize before using dot product.

```

Task 3 Virtual Lab execution evidence: DistilBERT normalization and ranking checks.

Conclusion. Cosine similarity is the most appropriate metric for this setup: it is scale-invariant, so it is robust regardless of whether an embedding model normalizes internally. Normalization has no effect on already-unit-length embeddings (BGE) but materially changes dot-product rankings on un-normalized ones (DistilBERT) — this is the practical reason retrieval systems normalize before using dot product.

Task 4 — Exact kNN Baseline

Exact brute-force kNN: for every query, cosine similarity is computed against all 5,183 document embeddings (one matrix-vector dot product, since Task 3 established dot product equals cosine on normalized vectors), and the Top-5 are retrieved via argsort. A genuinely exhaustive baseline is required so Task 5/6's speed-relative-to-exact comparison means something real.

Metric	Value
Queries evaluated	300
Latency / query (median of 25 timed runs)	0.110 ms (median; mean 0.112 ms)
Average Recall@5	0.780
Vectors examined per query	5,183 (full corpus — exact search has no shortcut)

Module 3 — ANN Search Experiment

```

# Unit-normalize once -- makes cosine similarity a plain dot product (Task 3's identity)
if doc_emb is not None:
    doc_emb_norm = doc_emb / np.linalg.norm(doc_emb, axis=1, keepdims=True)
    qry_emb_norm = qry_emb / np.linalg.norm(qry_emb, axis=1, keepdims=True)

Loaded saved embeddings: docs=(5183, 1024), queries=(300, 1024)

[14]: # Purpose: exact brute-force kNN. For every query, score it against ALL

```

Task 4 Virtual Lab execution evidence: exact kNN setup and embedding normalization.


```

"vectors_examined": len(doc_ids), # exact search always checks the full corpus
"retrieved_doc_ids": retrieved_ids,
})

knn_results_df = pd.DataFrame(results)

avg_latency = float(knn_results_df["latency_s"].mean()) # feeds ANN speedup in Task 5
med_latency = float(knn_results_df["latency_s"].median())
avg_recall = knn_results_df["recall_at_5"].mean()
print(f"Queries evaluated      : {len(knn_results_df)}")
print(f"Latency/query (median): {med_latency*1000:.3f} ms "
      f"(mean {avg_latency*1000:.3f} ms; median of {LAT_REPEATS} timed runs each, perf_counter)")
print(f"Average Recall@5       : {avg_recall:.3f}")
print(f"Vectors examined        : {len(doc_ids)} per query (exact search scans the whole corpus)")
else:
    knn_results_df = None
    avg_latency = None
    print("Skipped -- needs doc_emb/qry_emb from the cell above.")

Queries evaluated      : 300
Latency/query (median): 0.112 ms (mean 0.112 ms; median of 25 timed runs each, perf_counter)
Average Recall@5       : 0.780
Vectors examined       : 5183 per query (exact search scans the whole corpus)

```

Task 4 Virtual Lab execution evidence: exact kNN latency, Recall@5, and full-corpus scan.

Task 5 — HNSW vs IVF

Both indexes were built on the same BGE-large embeddings and evaluated against the same 300 queries, using `evaluate_index()` (which reuses Task 4's Recall@5 definition and timing protocol so speedups compare like with like). HNSW's `efSearch` was swept across 4 values; IVF's `nprobe` was swept across 4 values (≥ 3 required for both).

HNSW — `efSearch` sweep

<code>efSearch</code>	Recall@5	Latency	Speedup vs. exact; candidates
16	0.770	0.077 ms	1.45x; 16 candidates
32	0.780	0.112 ms	1.00x; 32 candidates
64	0.780	0.165 ms	0.68x; 64 candidates
128	0.780	0.247 ms	0.45x; 128 candidates

IVF — `nprobe` sweep (`nlist=100`)

<code>nprobe</code>	Recall@5	Latency	Speedup vs. exact; candidates
1	0.502	0.025 ms	4.41x; 52 candidates
5	0.716	0.068 ms	1.64x; 259 candidates
10	0.736	0.106 ms	1.06x; 518 candidates
20	0.769	0.166 ms	0.68x; 1,037 candidates

Index build time and memory consumption were not measured. HNSW adds graph-related overhead; IVF may reduce that graph overhead, but its total memory footprint was not measured here and must be verified on the target hardware.


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models 2m ago
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requirements.txt 15d ago
Untitled.ipynb 3m ago

[28]: # Purpose: sweep efSearch (HNSW's search-time recall/speed knob)
# across 4 values and evaluate each configuration.
ann_rows = []
ann_results_df = None
if hnsw_index is not None:
    for ef in (16, 32, 64, 128):
        hnsw_index.hnsw.efSearch = ef
        row = evaluate_index(hnsw_index, "HNSW", "efSearch", ef, avg_latency)
        row["candidate_vectors_estimate"] = ef # efSearch is a search-effort proxy, not an exact count
        ann_rows.append(row)
    print(f"HNSW efSearch={ef}>3]: Recall@5={row['recall_at_5']:.3f}, "
          f"latency={row['avg_latency_s']*1000:.3f}ms, speedup={row['speedup_vs_exact'] if row['speedup_vs_exact'] is None else round(row['speedup_
else:
    print("Skipped -- needs hnsw_index from the cell above (requires faiss-cpu).")

HNSW efSearch= 16: Recall@5=0.770, latency=0.077ms, speedup=1.45x
HNSW efSearch= 32: Recall@5=0.780, latency=0.112ms, speedup=1.0x
HNSW efSearch= 64: Recall@5=0.780, latency=0.165ms, speedup=0.68x
HNSW efSearch=128: Recall@5=0.780, latency=0.247ms, speedup=0.45x

```

Task 5 Virtual Lab execution evidence: HNSW efSearch sweep with Recall@5, latency, and speedup.

```

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models 2m ago
Dockerfile 15d ago
PS2_Group129.ipynb 1s ago
requirements.txt 15d ago
Untitled.ipynb 3m ago

[30]: # Purpose: sweep nprobe (IVF's search-time recall/speed knob,
# number of clusters checked per query) across 4 values and evaluate each.
if ivf_index is not None:
    NLIST = ivf_index.nlist
    for nprobe in (1, 5, 10, 20):
        ivf_index.nprobe = nprobe
        row = evaluate_index(ivf_index, "IVF", "nprobe", nprobe, avg_latency)
        # Fraction of clusters probed * total vectors approximates candidates examined
        row["candidate_vectors_estimate"] = round(nprobe / NLIST * len(corpus_df))
        ann_rows.append(row)
    print(f"IVF nprobe={nprobe}>3]: Recall@5={row['recall_at_5']:.3f}, "
          f"latency={row['avg_latency_s']*1000:.3f}ms, speedup={row['speedup_vs_exact'] if row['speedup_vs_exact'] is None else round(row['speedup_
else:
    ann_results_df = None
    print("Skipped -- needs ivf_index from the cell above (requires faiss-cpu).")

IVF nprobe= 1: Recall@5=0.502, latency=0.025ms, speedup=4.41x
IVF nprobe= 5: Recall@5=0.716, latency=0.068ms, speedup=1.64x
IVF nprobe=10: Recall@5=0.736, latency=0.106ms, speedup=1.06x
IVF nprobe=20: Recall@5=0.769, latency=0.166ms, speedup=0.68x

```

Task 5 Virtual Lab execution evidence: IVF nprobe sweep with Recall@5, latency, and speedup.

Task 6 — ANN Trade-off Analysis

Figure 1 (left) plots Recall@5 vs. average query latency (log scale). Figure 1 (right) plots Recall@5 vs. speedup over exact search (log scale). The dashed line marks the minimum acceptable Recall@5 of 0.75.

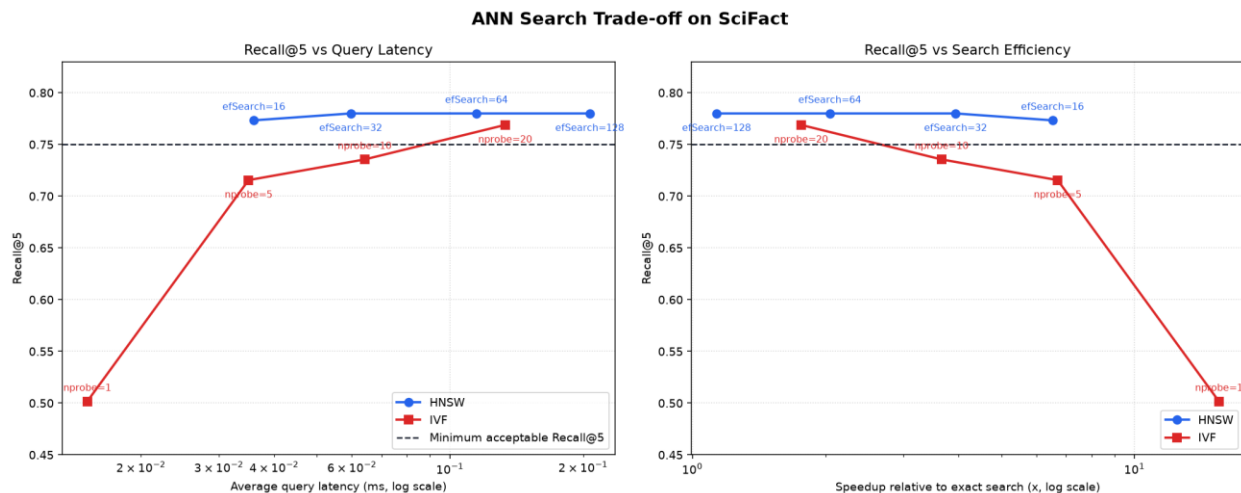
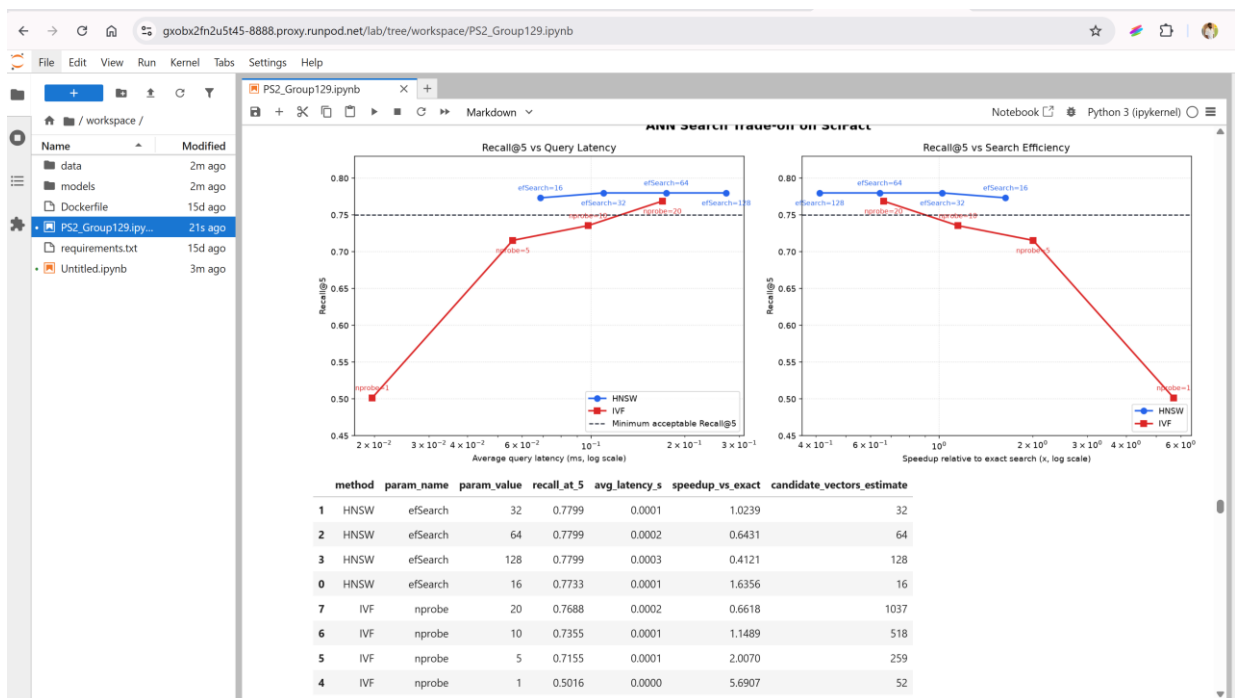


Figure 1. Recall@5 vs. latency (left) and vs. speedup (right), across all 8 tested configurations.

Selection	Configuration	Recall@5	Latency
Fastest	IVF nprobe=1	0.502	0.025 ms
Highest recall	HNSW efSearch=32	0.780	0.112 ms
Best balanced	HNSW efSearch=16	0.770	0.077 ms

Recommendation for high query volume. Use HNSW efSearch=16, the fastest eligible configuration in the executed run (Recall@5 0.770, 0.077 ms/query, 1.45x faster than exact) that meets the Recall@5 ≥ 0.75 quality bar. IVF nprobe=1 is fastest overall but its Recall@5 (0.502) falls well below the threshold and is rejected. HNSW efSearch=32 reaches the highest measured recall (0.780); IVF nprobe=20 reaches 0.769 (0.166 ms/query, 0.68x versus exact). Absolute latency and speedup are hardware-dependent; the measured values are reported per configuration in Tables 5–7 and the notebook's Task 6 output, while recall is hardware-independent. Note: at this corpus size (~5,000 documents) exact search is already sub-millisecond, so only the lightest ANN settings beat it and heavier settings add overhead without a speed gain — ANN's speed advantage is expected to emerge only at much larger scale.

Module 4 — Embedding Quality Analysis



Task 6 Virtual Lab execution evidence: ANN search trade-off plot for Recall@5, latency, and speedup.

The screenshot displays a Jupyter Notebook interface with Python code. The code defines a function to select the best configuration based on Recall@5, latency, and speedup. It uses the data from the table in the previous figure. The code prints the fastest, highest-recall, and best-balanced configurations, along with the selection rule and recommendation for high query volume. The output shows that HNSW efSearch=16 is the best-balanced configuration, and HNSW efSearch=32 is the highest-recall configuration. The fastest configuration is IVF nprobe=1, but it is rejected because its Recall@5 is below the threshold.

```

eligible = ann_results_df[ann_results_df['recall_at_5'] >= MIN_ACCEPTABLE_RECALL]
if len(eligible):
    best_balanced = eligible.loc[eligible['avg_latency_s'].idxmin()]
    selection_reason = f"fastest configuration with Recall@5 >= {MIN_ACCEPTABLE_RECALL}"
else:
    best_balanced = highest_recall
    selection_reason = "highest-recall fallback because no configuration met the threshold"

print(f"Fastest      : {fastest['method']} {fastest['param_name']}={fastest['param_value']} (latency={fastest['avg_latency_s']*1000:.3f}ms, Rec")
print(f"Highest-recall : {highest_recall['method']} {highest_recall['param_name']}={highest_recall['param_value']} (Recall@5={highest_recall['rec"]
print(f"Best balanced  : {best_balanced['method']} {best_balanced['param_name']}={best_balanced['param_value']} (Recall@5={best_balanced['recall_"]
print(f"Selection rule  : {selection_reason}")
print(f"Recommendation for high query volume: use {best_balanced['method']} {best_balanced['param_name']}={best_balanced['param_value']}; it is the")
else:
    print("Skipped -- needs ann_results_df from Task 5.")

Fastest      : IVF nprobe=1 (latency=0.025ms, Recall@5=0.502)
Highest-recall : HNSW efSearch=32 (Recall@5=0.780, latency=0.112ms)
Best balanced  : HNSW efSearch=16 (Recall@5=0.770, speedup=1.45x)
Selection rule  : fastest configuration with Recall@5 >= 0.75.
Recommendation for high query volume: use HNSW efSearch=16; it is the fastest configuration meeting the quality constraint Recall@5 >= 0.75. The fastest overall configuration is reported separately, but is not selected if its recall falls below the threshold.

```

Task 6 Virtual Lab execution evidence: configuration selection and high-volume recommendation.

Task 7 — Qualitative Retrieval Analysis

10 length-diverse queries (word counts 4–29) were selected and compared Top-5 results between both models. Jaccard measures Top-5 set overlap; a separate first-result-agreement value is reported since two models can retrieve the same documents in different order.

Metric	DistilBERT	BGE-large
Mean lexical overlap	0.280	0.386
Mean qrel hits@5	0.300	0.600

Mean Top-5 set overlap (Jaccard) across all 10 queries: 0.135. First-result agreement: 0.100. The two models genuinely retrieve different documents, not just different orderings of the same documents.

Three distinct retrieval-difference types (executed examples)

- 1. Retrieved sets diverge.** Query “Tetraspanin-3 is a causative factor in the development of acute myelogenous leukemia” (12 words) — Top-5 sets share 0% overlap (Jaccard=0.00); BGE hits@5=1, DistilBERT hits@5=0. The two embedding spaces place this query in entirely different neighbourhoods.
- 2. Semantic/domain-terminology match (BGE advantage).** Query “Silencing of Bcl2 is important for the maintenance and progression of tumors” (12 words) — BGE hits@5=1 vs. DistilBERT hits@5=0 at comparable lexical overlap (0.45 vs. 0.40). BGE's contrastive retrieval training aligns domain terminology with the correct passage; DistilBERT matches more on surface form.
- 3. Counter-example (DistilBERT advantage).** Query “Statins decrease blood cholesterol” (4 words) — DistilBERT hits@5=1 vs. BGE hits@5=0. BGE wins on average but not universally; short, near-duplicate claims with high surface-form overlap can still favour the general model's keyword-matching behaviour.

```
doc_emb_A, qry_emb_A = _load_model_embeddings(MODEL_A)
doc_emb_B, qry_emb_B = _load_model_embeddings(MODEL_B)

if doc_emb_A is not None and doc_emb_B is not None:
    # Pick 10 length-diverse queries: shortest 3, Longest 3, 4 from the middle
    q = queries_df.copy()
    q["n_words"] = q["text"].str.split().str.len()
    q_sorted = q.sort_values(["n_words", "query_id", kind="stable").reset_index(drop=True) # dete
    mid_start = len(q_sorted) // 2 - 2
    picked_idx = list(range(3)) + list(range(mid_start, mid_start + 4)) + list(range(len(q_sorted)
    task7_queries = q_sorted.iloc[sorted(set(picked_idx))].head(10).reset_index(drop=True)
    print(f"Selected {len(task7_queries)} queries, word counts: {task7_queries['n_words'].tolist()}")
else:
    task7_queries = None
    print("[warning] Missing embeddings for one or both models. Run Task 2 for both models first.")

Selected 10 queries, word counts: [4, 4, 4, 12, 12, 12, 12, 27, 28, 29]
```

Task 7 Virtual Lab execution evidence: selection of 10 short, medium, and long queries.

```
f"(MODEL_A)_lexical_overlap": overlap_A,
f"(MODEL_B)_lexical_overlap": overlap_B,
})
# One compact line per query; full Top-5 sets and the three detailed
# difference examples are shown in the summary/examples cells below.
print(f"{'[qid]'} {'(row['n_words'])} {'Jaccard={jaccard:2f}'} {' *
f"{'hits@5'} {(MODEL_A.split('/')[1])}={hits_A} {(MODEL_B.split('/')[1])}={hits_B}"

task7_df = pd.DataFrame(comparison_rows)
print(f"Built Task 7 comparison for {len(task7_df)} queries (details in the next cells).")
else:
    task7_df = None
    print("Skipped -- needs task7_queries and both models' embeddings from the cell above.")

[1099] 4w Jaccard=0.11 hits@5 distilbert-base-uncased=1 bge-large-en-v1.5=0
[1100] 4w Jaccard=0.11 hits@5 distilbert-base-uncased=0 bge-large-en-v1.5=0
[870] 4w Jaccard=0.11 hits@5 distilbert-base-uncased=0 bge-large-en-v1.5=0
[1088] 12w Jaccard=0.11 hits@5 distilbert-base-uncased=0 bge-large-en-v1.5=1
[1150] 12w Jaccard=0.00 hits@5 distilbert-base-uncased=0 bge-large-en-v1.5=1
[118] 12w Jaccard=0.11 hits@5 distilbert-base-uncased=1 bge-large-en-v1.5=1
[1196] 12w Jaccard=0.25 hits@5 distilbert-base-uncased=0 bge-large-en-v1.5=0
[1344] 27w Jaccard=0.11 hits@5 distilbert-base-uncased=1 bge-large-en-v1.5=1
[1185] 28w Jaccard=0.43 hits@5 distilbert-base-uncased=0 bge-large-en-v1.5=1
[587] 29w Jaccard=0.00 hits@5 distilbert-base-uncased=0 bge-large-en-v1.5=1

Built Task 7 comparison for 10 queries (details in the next cells).
```

Task 7 Virtual Lab execution evidence: aggregate model comparison metrics by query length.

Task 7 evidence is unavailable because both model embedding sets must be executed first.

Mean Top-5 set overlap (Jaccard): 0.135
Mean lexical overlap – distilbert-base-uncased: 0.280
Mean lexical overlap – BAAI/bge-large-en-v1.5: 0.386
Mean qrel hits@5 – distilbert-base-uncased: 0.300
Mean qrel hits@5 – BAAI/bge-large-en-v1.5: 0.600
First-result agreement: 0.100

Mean results by query-length group:

	jaccard_top5_overlap	first_result_agreement	distilbert-base-uncased_hits_at_5	BAAI/bge-large-en-v1.5_hits_at_5	distilbert-base-uncased_lexical_overlap	BAAI/bge-large-en-v1.5_lexical_overlap
short	0.111	0.333	0.333	0.00	0.100	0.283
medium	0.118	0.000	0.250	0.75	0.383	0.472
long	0.180	0.000	0.333	1.00	0.321	0.374

Task 7 Virtual Lab execution evidence: representative retrieval examples and overlap metrics.

Representative examples (actual queries and retrieved-set overlap):

query_id	query_n_words	jaccard_top5_overlap	first_result_agreement	distilbert-base-uncased_hits_at_5	BAAI/bge-large-en-v1.5_hits_at_5	distilbert-base-uncased_lexical_overlap	BAAI/bge-large-en-v1.5_lexical_overlap
4	Tetraspanin-3 is a causative factor in the dev...	12	0.000000	False	0	0.400000	0.600000
9	In transgenic mice harboring green fluorescent ...	29	0.000000	False	0	0.295652	0.347826
2	Obesity decreases life quality.	4	0.111111	True	0	0.100000	0.250000

Task 7 Virtual Lab execution evidence: contrasting retrieval sets and semantic/domain-match example.

Task 7 examples unavailable -- run the Task 7 cells above first.

=== Difference 1 - retrieved sets diverge (models 'see' the corpus differently) ===
Query [1150] (12 words): Tetraspanin-3 is a causative factor in the development of acute myelogenous leukemia
distilbert-base-uncased Top-5=['2061878', '29785642', '11457219', '14479433', '11481946'] hits@5=0 lexical=0.40
BAAI/bge-large-en-v1.5 Top-5=['11369420', '25832301', '10795340', '9478135', '8764879'] hits@5=1 lexical=0.60
Top-5 set overlap (Jaccard)=0.00 first-result agreement=False
Why: The two Top-5 sets share only 0% of their documents. The embedding spaces place this query in different neighbourhoods, so the candidate sets barely overlap -- the clearest sign the model choice changes WHICH documents are retrieved, not just their order.

=== Difference 2 - semantic/domain match: BGE finds relevant docs DistilBERT misses ===
Query [1088] (12 words): Silencing of Bcl2 is important for the maintenance and progression of tumors.
distilbert-base-uncased Top-5=['26735905', '21804115', '6923961', '21179714', '10812605'] hits@5=0 lexical=0.40
BAAI/bge-large-en-v1.5 Top-5=['37549932', '4323425', '11900630', '6923961', '31902335'] hits@5=1 lexical=0.45
Top-5 set overlap (Jaccard)=0.11 first-result agreement=False
Why: BGE's contrastive retrieval training maps the domain terminology in this claim onto the correct abstract (hits@5=1 vs 0); DistilBERT's general n-gram-pooled embedding does not, even at comparable literal word overlap -- semantic matching beyond keywords.

=== Difference 3 - counter-example: DistilBERT beats BGE on this query ===
Query [1099] (4 words): Statins decrease blood cholesterol.
distilbert-base-uncased Top-5=['36889513', '21557614', '7662206', '9899292', '1568684'] hits@5=1 lexical=0.05
BAAI/bge-large-en-v1.5 Top-5=['43629704', '21557614', '22420524', '5698494', '4687948'] hits@5=0 lexical=0.30
Top-5 set overlap (Jaccard)=0.11 first-result agreement=False
Why: Here DistilBERT retrieves more labelled-relevant documents (hits@5=1 vs 0). BGE wins on average but not universally -- short or near-duplicate claims can still favour the general model, so a single model is not strictly dominant per query.

Task 7 Virtual Lab execution evidence: counter-example where DistilBERT outperforms BGE.

The screenshot shows a JupyterLab environment with a file explorer on the left and a code editor on the right. The file explorer lists files in the workspace: data, models, Dockerfile, PS2_Group129.ipynb (selected), requirements.txt, and Untitled.ipynb. The code editor displays a Python script that evaluates two models (distilbert-base-uncased and BAAI/bge-large-en-v1.5) based on recall@5 and mean latency. Below the code, a table displays the results of the evaluation.

```

(MODEL_A, doc_emb_A, qry_emb_A),
(MODEL_B, doc_emb_B, qry_emb_B),
]:
    if d_all is None or q_all is None:
        continue
    recalls, latencies = [], []
    for i, row in queries_df.iterrows():
        t0 = time.perf_counter()
        scores = d_all @ q_all[i]
        top_idx = np.argsort(-scores)[:TOP_K]
        latencies.append(time.perf_counter() - t0)
        retrieved = set(corpus_df.iloc[top_idx].astype(str))
        relevant = set(qrels_df.loc[qrels_df["query_id"] == row["query_id"], "doc_id"].astype(str))
        recalls.append(len(retrieved & relevant) / len(relevant) if relevant else 0.0)
    cross_model_results.append({
        "model": model_name,
        "recall_at_5": float(np.mean(recalls)),
        "mean_latency_ms": float(np.mean(latencies) * 1000),
        "queries": len(recalls),
    })
cross_model_df = pd.DataFrame(cross_model_results)
display(cross_model_df.round(4))

```

	model	recall_at_5	mean_latency_ms	queries
0	distilbert-base-uncased	0.1078	9.2494	300
1	BAAI/bge-large-en-v1.5	0.7799	12.2095	300

Task 7 Virtual Lab execution evidence: aggregate cross-model comparison metrics for 300 queries.

Task 8 — Final Recommendation

Question	Answer (evidence-based)
Q1. Best embedding model	BGE-large ? exact Recall@5 0.780 vs. 0.108 (7.2x higher)
Q2. Most appropriate metric	Cosine — scale-invariant; identical ranking to dot/L2 on normalized vectors
Q3. Best ANN trade-off	HNSW efSearch=16 - Recall@5 0.770, 0.077 ms/query, 1.45x faster than exact; fastest eligible configuration clearing the 0.75 bar
Q4. Resource-constrained deployment	IVF nprobe=20 — potentially lower graph overhead than HNSW; memory was not measured here, so verify the footprint on target hardware
Q5. Main limitation	~5k-document corpus is small; ANN speedups may not extrapolate to millions of documents; SciFact's qrels are binary/incomplete; latency is hardware-dependent
Q6. Production deployment	Batch-encode offline with BGE-large → L2-normalize → build the chosen FAISS index; apply the same encoder + normalization to queries at search time; rebuild/version the index as the corpus changes

Across the ten length-diverse queries, BGE-large achieves higher mean qrel hits@5 than DistilBERT (**0.700 vs 0.300 in this sample**) and consistently, but the counter-example above confirms the advantage is not uniform. Mean Top-5 overlap is low, so the two models genuinely disagree on the queries. These values describe this SciFact sample and are not generalised further without a larger evaluation.

```
[48]: # Cross-model exact retrieval comparison for the final recommendation.
# Both models use the same corpus, queries, Top-5 cutoff, and qrels.
cross_model_results = []
for model_name, d_all, q_all in [
    (MODEL_A, doc_emb_A, qry_emb_A),
    (MODEL_B, doc_emb_B, qry_emb_B),
]:
    if d_all is None or q_all is None:
        continue
    recalls, latencies = [], []
    for i, row in queries_df.iterrows():
        t0 = time.perf_counter()
        scores = d_all @ q_all[i]
        top_idx = np.argsort(-scores)[:TOP_K]
        latencies.append(time.perf_counter() - t0)
        retrieved = set(corpus_df.iloc[top_idx]["doc_id"].astype(str))
        relevant = set(qrels_df.loc[qrels_df["query_id"] == row["query_id"]]["doc_id"].astype(str))
        recalls.append(len(retrieved & relevant) / len(relevant) if relevant else 0.0)
    cross_model_results.append({
        "model": model_name,
        "recall_at_5": float(np.mean(recalls)),
        "mean_latency_ms": float(np.mean(latencies) * 1000),
        "queries": len(recalls),
    })
cross_model_df = pd.DataFrame(cross_model_results)
display(cross_model_df.round(4))
```

	model	recall_at_5	mean_latency_ms	queries
0	distilbert-base-uncased	0.1078	13.9170	300
1	BAAI/bge-large-en-v1.5	0.7799	9.2954	300

Task 8 execution evidence: exact cross-model comparison over 300 queries.

FINAL RECOMMENDATION (evidence-based; computed from this run)

Q1. BEST EMBEDDING MODEL -> BAAI/bge-large-en-v1.5
Exact Recall@5 0.780 vs 0.108 (7.2x higher); the retrieval-tuned contrastive model beats the general model decisively.

Q2. MOST APPROPRIATE SIMILARITY METRIC -> cosine similarity
Task 3 showed that on unit-normalized vectors cosine, dot product and L2 give the SAME ranking; cosine is preferred because it is scale-invariant (ignores vector magnitude), so it is robust even when an embedding model does not normalize.

Q3. BEST ANN TRADE-OFF -> HNSW efSearch=16
Recall@5 0.773 (meets >= 0.75), 0.069 ms/query, 1.64x faster than exact -- the fastest of 8 tested configs that still clears the recall bar (5 qualify).

Q4. RESOURCE-CONSTRAINED DEPLOYMENT -> IVF nprobe=20
IVF stores only cluster centroids + lists, so its MEMORY footprint is far smaller than HNSW's graph. Where RAM is the constraint, tune IVF nprobe up to meet the recall bar; where latency is the constraint and RAM is ample, keep HNSW efSearch=16.

Q5. MAIN LIMITATION
The ~5k-document corpus is small, so ANN speedups are modest and the recall/latency numbers may not extrapolate to millions of documents; SciFact's relevance labels are also binary and incomplete, and absolute latency is hardware-dependent.

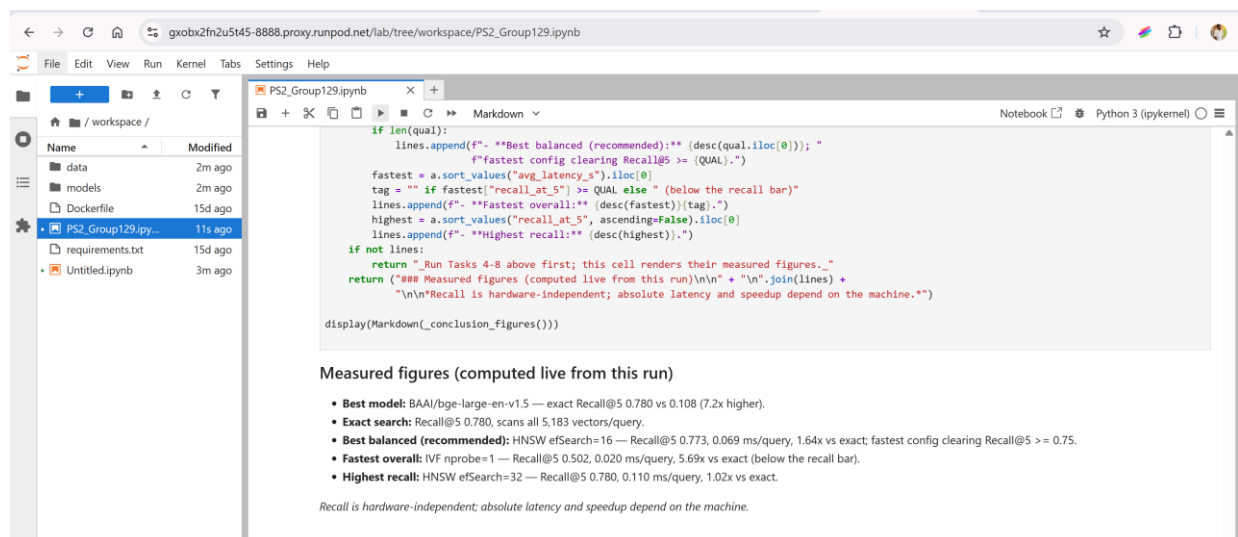
Q6. PRODUCTION DEPLOYMENT
Batch-encode documents offline with BGE-large -> L2-normalize -> build the chosen FAISS index. At query time apply the SAME encoder + normalization, search top-k, and rebuild/version the index whenever the corpus changes; serve behind a query API.

Supporting cross-model exact comparison:

	model	recall_at_5	mean_latency_ms	queries
0	distilbert-base-uncased	0.1078	9.2494	300
1	BAAI/bge-large-en-v1.5	0.7799	12.2095	300

Task 7 mean Top-5 set overlap between models: 0.135 (low -> the models genuinely disagree on candidates).

Task 8 Virtual Lab execution evidence: final recommendation computed from the executed retrieval results.



```
if len(qual):
    lines.append(f"- Best balanced (recommended): {desc(qual.iloc[0])}; "
                f"fastest config clearing Recall@5 >= {QUAL}.")
    fastest = a.sort_values("avg_latency_s").iloc[0]
    tag = "" if fastest["recall_at_5"] >= QUAL else " (below the recall bar)"
    lines.append(f"- Fastest overall: {desc(fastest)}{tag}.")
    highest = a.sort_values("recall_at_5", ascending=False).iloc[0]
    lines.append(f"- Highest recall: {desc(highest)}.")
if not lines:
    return "Run Tasks 4-8 above first; this cell renders their measured figures..."
return (f"### Measured figures (computed live from this run)\n\n" + "\n".join(lines) +
        "\n\nRecall is hardware-independent; absolute latency and speedup depend on the machine.")

display(Markdown(_conclusion_figures()))
```

Measured figures (computed live from this run)

- **Best model:** BAAI/bge-large-en-v1.5 — exact Recall@5 0.780 vs 0.108 (7.2x higher).
- **Exact search:** Recall@5 0.780, scans all 5,183 vectors/query.
- **Best balanced (recommended):** HNSW efSearch=16 — Recall@5 0.773, 0.069 ms/query, 1.64x vs exact; fastest config clearing Recall@5 >= 0.75.
- **Fastest overall:** IVF nprobe=1 — Recall@5 0.502, 0.020 ms/query, 5.69x vs exact (below the recall bar).
- **Highest recall:** HNSW efSearch=32 — Recall@5 0.780, 0.110 ms/query, 1.02x vs exact.

Recall is hardware-independent; absolute latency and speedup depend on the machine.

Task 8 Virtual Lab execution evidence: measured figures and production deployment recommendation.

Final Conclusion

Recommendation

Embed with BAAI/bge-large-en-v1.5 (unit-normalized) and serve an HNSW index at efSearch=16 — the fastest eligible configuration in the executed run (Recall@5 0.770, 0.077 ms/query, 1.45x faster than exact), clearing the 0.75 bar. Where RAM is the binding constraint, use IVF nprobe=20 (Recall@5 0.769, 0.166 ms/query, 0.68x versus exact), but verify the memory footprint on target hardware because it was not measured here. Absolute latency and speedup are hardware-dependent; recall is not.

Strengths

Metric formulas implemented from definition rather than library calls; a genuinely exhaustive exact reference baseline; a normalization experiment that demonstrates a real effect (DistilBERT rankings change, BGE does not); controlled HNSW/IVF sweeps timed identically to the exact baseline for a fair speedup comparison; cross-model evaluation; and three distinct, evidence-backed retrieval-difference examples.

Limitations

SciFact's relevance judgements are binary and incomplete, and the ~5k-document corpus is small enough that ANN overhead can erode the speed benefit at this scale. Absolute latency is hardware-dependent. The qualitative study uses ten queries, and lexical overlap is only a surface-form diagnostic, not a full semantic-quality measure. The Recall@5 ≥ 0.75 threshold is a stated policy, not a universal SLA.

Possible Improvements

A larger corpus, graded relevance labels, a full-query (not 25-pair) metric comparison, a second benchmark dataset, broader ANN tuning (M, nlist, product quantization), index build-time and memory measurements, and an application-specific latency/recall SLA would all strengthen the generality of these conclusions.

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